Biochemistry Cume
February 11, 2005

Protein Structure

1. What are the PrP^c (cellular) and PrP^{sc} (Scrapie) forms of prion protein? Explain why they have different biological functions even they share the same amino acid sequence. Provide your comments concerning protein structure prediction within the context of this example. (25 %)

2. Myoglobin contains 8 alpha helices, one of which has the following sequence:

   QGAMN KALEH FRKDI AA

   Which side chains are likely to be on the side of the helix that faces the interior of the protein? Which are likely to be facing the aqueous solvent? Account for the spacing of the residues facing the interior. (25 %)

3. Fetal hemoglobin (Hb F) contains serine in place of cationic histidine 143 of the beta chain of adult hemoglobin (Hb A). Residue 143 faces the central cavity between the beta chains.

a) Why does 2,3-bisphosphoglycerate (2,3-BPG) bind more tightly to deoxy Hb A than to deoxy Hb F? (8 %)

b) How does the decrease of the affinity of Hb F for 2,3-BPG affect the affinity of Hb F for O_2? (8 %)

c) The P_{50} of Hb F is 18 torr, anf the P_{50} of Hb A is 26 torr. How do these values explain the efficient transfer of the oxygen from maternal blood to fetus? (9 %)

4. Circular dichroism measurements have shown that pol-L-lysine is a random coil at pH=7, but becomes alpha-helical as the pH is raised above 10. Account for this pH-dependent conformational transition. Also predict the pH dependence of the helix-coil transition of poly-L-glutamate. (25 %)

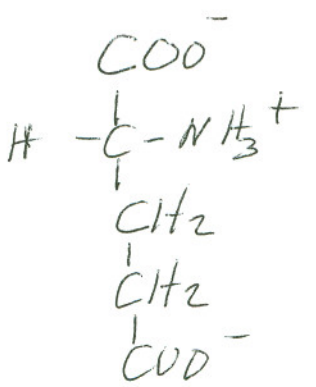
Biochemistry Cumulative Exam
March 4, 2005

1. (40) Can glutamate be used as the complete carbon source and nitrogen source for the synthesis of aspartate in a liver cell that is not supplied with carbohydrate or other amino acids? If yes, than give a complete metabolic pathway for the synthesis of aspartate from glutamate. Also, give a complete balanced equation for the reaction giving the stoichiometry of all reactants and products.

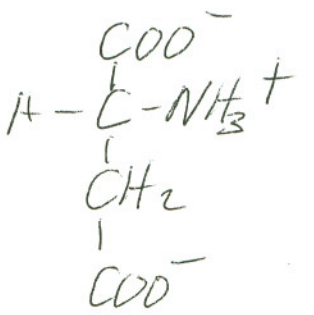
2. (30) If glutamate radiolabeled at carbon 2 is added to the liver cell under the above conditions, which carbon atom(s) of aspartate will first become labeled? Show your work

3. (30) If glutamate radiolabeled at carbon 2 is added to the liver cell under the above conditions, which carbon atom(s) of alanine and glucose will first become labeled? Give metabolic pathways for these reactions, and show your work.

(Supplementary Information Attached.)



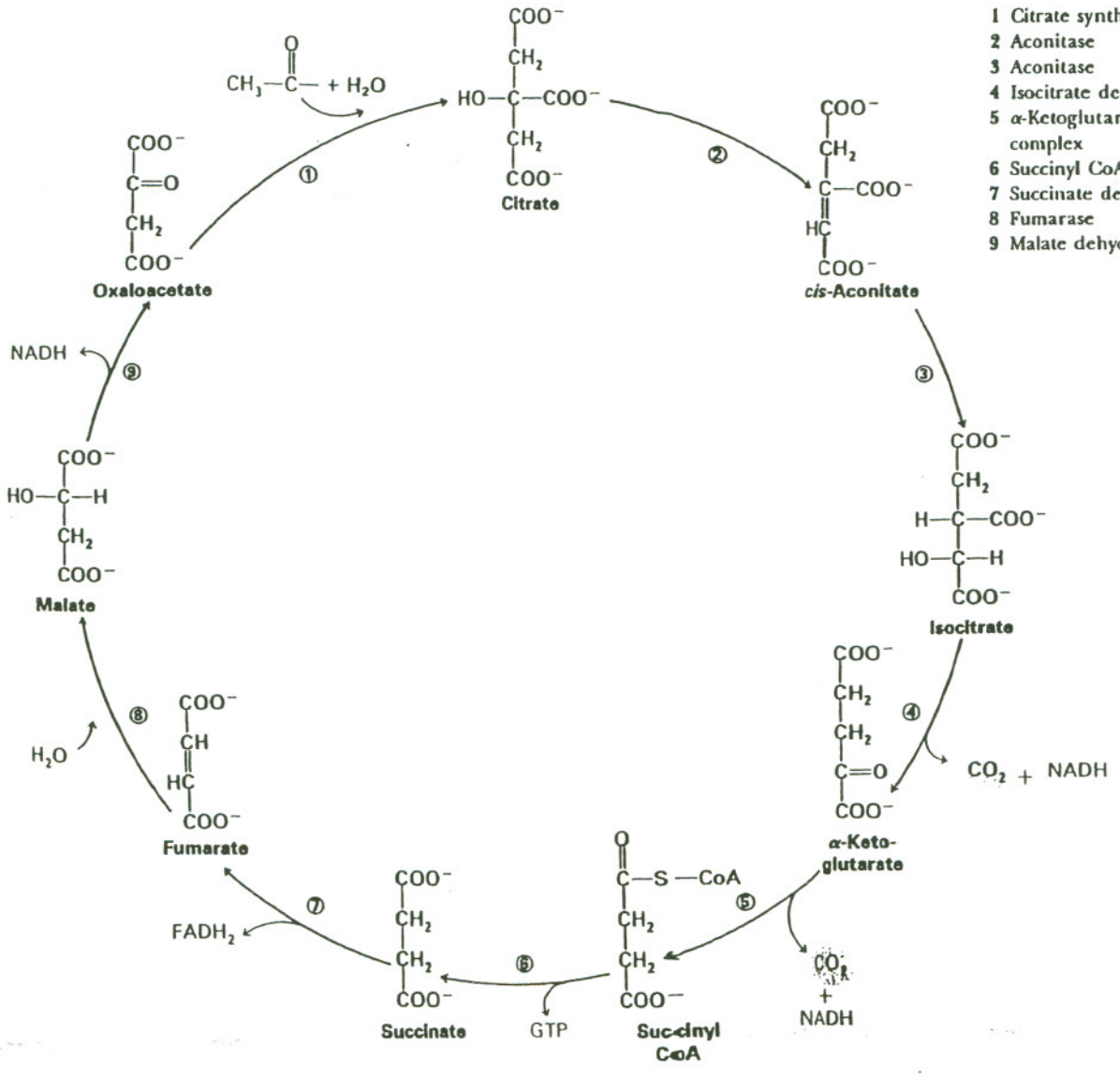
Glutamate

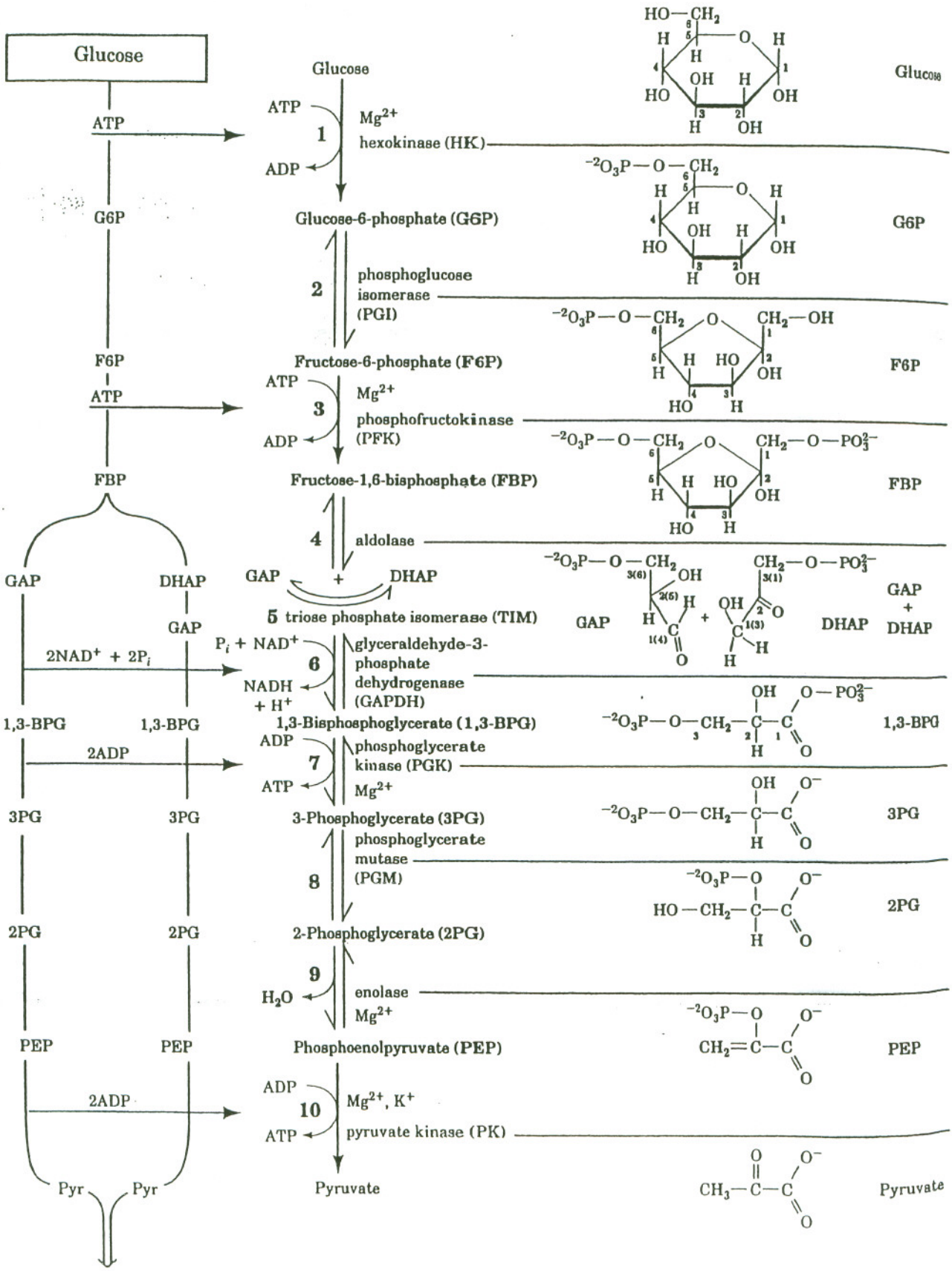


Aspartate

following enzymes, as numbered in the diagram:

1 Citrate synthase
2 Aconitase
3 Aconitase
4 Isocitrate dehydrogenase
5 α-Ketoglutarate dehydrogenase complex
6 Succinyl CoA synthetase
7 Succinate dehydrogenase
8 Fumarase
9 Malate dehydrogenase




Glucose
ATP
G6P
F6P
ATP
FBP
GAP
DHAP
GAP
2NAD+ + 2Pi
1,3-BPG
1,3-BPG
2ADP
3PG
3PG
2PG
2PG
PEP
PEP
2ADP
Pyr
Pyr
Glucose
ATP
1
Mg2+
hexokinase (HK)
ADP
Glucose-6-phosphate (G6P)
2
phosphoglucose isomerase (PGI)
Fructose-6-phosphate (F6P)
ATP
3
Mg2+
phosphofructokinase (PFK)
ADP
Fructose-1,6-bisphosphate (FBP)
4
aldolase
GAP + DHAP
5 triose phosphate isomerase (TIM)
Pi + NAD+
6
glyceraldehyde-3-phosphate dehydrogenase (GAPDH)
NADH + H+
1,3-Bisphosphoglycerate (1,3-BPG)
ADP
7
phosphoglycerate kinase (PGK)
ATP
Mg2+
3-Phosphoglycerate (3PG)
8
phosphoglycerate mutase (PGM)
2-Phosphoglycerate (2PG)
9
H2O
enolase
Mg2+
Phosphoenolpyruvate (PEP)
ADP
10
Mg2+, K+
pyruvate kinase (PK)
ATP
Pyruvate
Glucose
G6P
F6P
FBP
GAP
DHAP
GAP + DHAP
1,3-BPG
3PG
2PG
PEP
Pyruvate

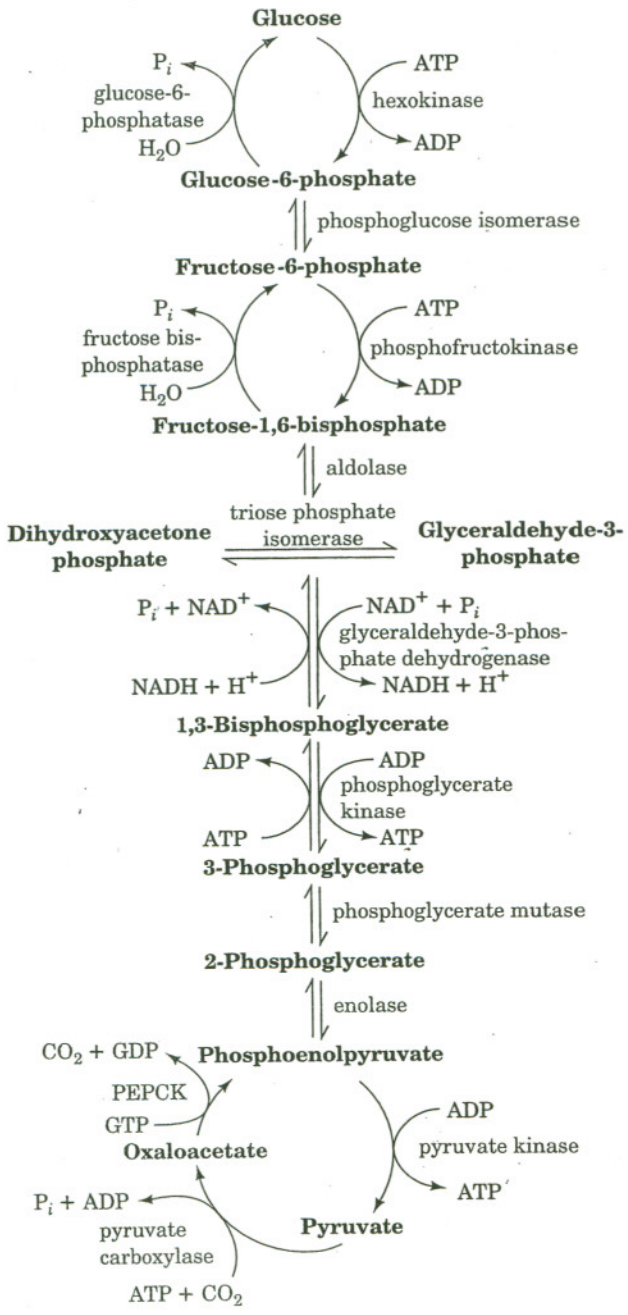


Figure 15-22. Key to Metabolism. **Comparison of the pathways of gluconeogenesis and glycolysis.** The red arrows represent the steps that are catalyzed by different enzymes in gluconeogenesis. The other seven reaction steps of gluconeogenesis are catalyzed by glycolytic enzymes that function near equilibrium.